

DBA5101– Analytics in Managerial Economics

Group Project 1

Group 11

Chew Hong Jin (A0239059W)

Kim Joohee (A0319398L)

Udhaya Rathinavel (A0319388M)

Sandra Lim (A0318020X)

Sheryl Lim (A0318804A)

AI Tool Declaration

We used chatGPT5 to support the preparation of this assignment. Specifically, the tool was used to structure answers, refine the language, improve clarity of expression, and correct coding. All outputs from the AI were reviewed, edited, and integrated by group. Reference to Stack overflow was made. We take full responsibility for the content, analysis, and quality of the submitted work.

1. Introduction

Understanding how passengers respond to ticket prices and attributes enables operators to design policies that balance profitability with service accessibility.

In particular, we examine whether purchase demand is systematically influenced by price, advance purchase days and ticket type.

2. Data Observation

- Dataset: Transaction-level sales data from a major train station.
- Key variables: number of tickets sold, mean net price, purchase vs. departure date, ticket type, cabin type, customer category, train identifiers.
- Initial insights:
 - Inverse relationship between price and quantity demanded (*Refer to Appendix Figure 2.1A and 2.1B*)
 - Wide variation in advance purchase days . (*Refer to Appendix Figure 2.2*)
 - Seasonality (Mar/Jun/Dec peaks reflective of holiday periods). (*Refer to Appendix Figure 2.3A, 2.3B*)
 - Presence of train effects is evident, as some train numbers consistently sell significantly more tickets than others, even across different months (*Refer to Appendix Figure 2.4 - 2.5*)

3. Data Cleaning

- Checked for duplicates and corrected inconsistent date fields.
- Generated $\text{Advance_days} = \text{Dept_Date} - \text{Purchase_Date}$ for potential endogeneity checking
- Generated additional variables to handle missing/quirky flags (e.g., ticket_type as derived variable to encode joint interaction of isReturn and isOneway tickets)
- Dropped extreme outliers in prices using **Tukey's IQR rule** (*Refer to Appendix Table 3.1*)

4. Regression function (OLS) and Interpretation

Levels demand function: $Q = \alpha + \beta P + \beta_1 X_1 + \beta_2 X_2 + \dots + \varepsilon$

- Q : seats sold
- P : average ticket price
- X_i : Key variables (Ticket Type, Departure Month, customer category, Train number, Cabin Type)
- β : the coefficient of interest
- Limitation: direct interpretation of β is in “units of seats per currency unit,” which is less intuitive.

Created log-transformed variables to normalise skewed data and reduce outlier effects: Elasticities like Price Elasticity of Demand are unit-free, so the log form enables meaningful comparisons across markets or cabin classes. It also stabilizes variance, captures proportional (not absolute) changes, and reduces the impact of skewed or extreme values in sales and prices.

Log - Log demand function: $\ln(Q) = \alpha + \beta \ln(P) + \beta_1 X_1 + \beta_2 X_2 + \dots + \varepsilon$

- $\ln(Q)$: log of seats sold (dependent variable)
- $\ln(P)$: log of mean ticket price

Baseline OLS Regression	Baseline Levels Model (Appendix Figure 4.1)	Baseline Log Model (Appendix Figure 4.2)
Dependent Variable	num_seats_total	Log(num_seats_total)
Ticket	-0.2668 (0.017)	-0.1045 (0.005)
Ticket Type_Both	-6.992e-05 (0.025)	-0.0028(0.007)
Ticket Type_Neither	0.0315 (0.015)	0.0065 (0.005)
Departure Month Feb - Dec	Range from -0.1015 to 0.6663 (Range from 0.019 to 0.022)	Range from -0.0546 to 0.2347 (Range from 0.006 to 0.007)
Customer Category	0.7047 (0.009)	0.2716 (0.003)
Train Number B - O	Range from -1.2836 to 0.6161 (Range from 0.021 to 235.875)	Range from -0.5551 to 0.2004 (Range from 0.007 to 19.333)
Mean_net_ticket_price	-0.0008 (4.61e-05)	-0.1929 (0.004)
Days in Advance	0.0025 (9.14e-05)	0.0005 (2.75e-05)
Cabin Type	0.0539 (0.012)	-0.0501 (0.004)
Observations	209,694	209,694
R²	0.097	0.147

- This table reports results from an OLS regression of number of seats sold on ticket type, departure month, customer category, train number, mean net ticket price, days in advance and whether the seat is for normal cabin.
- Results for Departure Month and Train Number have been summarised to show min and max as there are too many categorical variables.
- Multi-collinearity was checked using correlation matrix and VIF statistics. All VIF values were below 5, suggesting that collinearity is not a concern in our specification (*Refer to Appendix Figure 4.3*).

5. Test Hypothesis & Interpretation

- **Key coefficient** ($\ln_P$): **-0.19** ($p < 0.01$). This means a 1% increase in ticket price reduces ticket sales by ~0.19%. **Demand is relatively price inelastic.**
- T-Test: H_0 : Ticket price has no effect ($\beta = 0$). H_1 : Higher prices reduce demand ($\beta < 0$). OLS gives $\beta = -0.19$ ($p < 0.01$), with $t > 1.96$. Reject H_0 : higher prices significantly reduce demand, though demand is relatively inelastic.
- Cabin Type: Negative, suggesting premium cabins may sell fewer seats.
- Departure Month: Clear seasonality (higher demand in selected months relative to January, such as March and June).
- Customer Category: Positive, indicating business customers book more seats.
- Train number: Important, as some trains systematically sell more seats.
- Issue: While demand is relatively price inelastic, this estimate may be biased because of the dynamic pricing strategy seen in the data (*Refer to Appendix Figure 4.4*). Prices vary significantly and tapers down as the gap between purchase and departure dates decrease..

6. Endogeneity Problem and Instrumental Variables

- Endogeneity Problem
Ticket price and demand are jointly determined through dynamic pricing, which creates **simultaneity bias in OLS**.
- Instrumental Variable Solution
We use **Advance_days** as an IV. It meets two conditions:
1) Relevance: Strongly correlated with price ($\text{corr} \approx -0.51$) due to fare rules (*Refer to Appendix Figure 4.3 for data*). 2) Exogeneity: Affects demand only via price, not through unobserved shocks.
Thus, we believe **Advance_days** is a strong and valid instrument for price.
- The strong first stage fit (R^2) and highly significant F-statistic (regardless of the levels or log model) confirm that **Advance_days** is a good predictor of ticket price, supporting its validity as an instrument. (*Refer to Table below for data*)

	2SLS - Levels	2SLS - Log
First Stage Estimation		
R²	0.6214	0.6556
Partial F-Statistic	2.238e+04	2.66e+04
P-Value	0.000	0.0000
Hausman Endogeneity Test		
F-Statistic	1.003e+05	2.248e+05
P-Value	0.0000	0.0000

- **Hausman test** confirms : The significant **Hausman test** statistic allows us to reject the null hypothesis of exogeneity, confirming that ticket price is endogenous and that OLS estimates are biased, making the 2SLS the preferred estimator of the demand function.

7. Final 2SLS Formula with IV

$$\ln(Q) = \alpha + \beta \ln(P) + \beta_G X_G + \beta_k X_k + \dots + \varepsilon$$

$$\ln(P) = \beta_G X_G + v$$

- β : the price elasticity of demand.
- X_k : vector of controls (cabin type, ticket type, customer category, train number)
- β_i the coefficient of variables
- X_G : Advance days (IV)
- ε : error term
- v : first stage error term

8. Comparison 2SLS vs. OLS

Second Stage SLS	OLS – Levels (Figure 4.1)	OLS-Log (Figure 4.2)	2SLS – Levels (Figure 5.1C)	2SLS – Log (Figure 5.2C)
Coefficient	-0.0008	-0.1929	-0.0051	-0.3747
Standard Error	4.61e-05	0.004	0.0001	0.0086
R ²	0.097	0.147	0.0636	0.1376

- Based on the 2SLS, price elasticity of demand, although relatively low, is twice that of the OLS (-0.19 → -0.37).
- This further confirms that the OLS is biased and underestimated true price sensitivity due to the endogeneity.
- Although the R² based on the 2SLS (0.1376) is lower than the OLS (0.147), this is not unexpected as there is a trade-off between bias and variance. As such, while there is reduction in fit of the model, the estimates are more reliable.
- **2SLS provides more reliable estimates**, indicating that ticket demand is still relatively more price-sensitive than what is shown in the OLS, albeit still inelastic.
- Managerial Insights: Operators should adopt dynamic, segment-based pricing by offering early-bird discounts for price-sensitive travellers, adjusting fares for peak/holiday seasons, and differentiating by cabin—economy promotions vs. premium perks for first-class.